Modelling process :

Data Collection → Database, opensource datasets, kaggle ....

⇓

EDA (Understand distributions) ⇒ Continous processes

EDA 1
EDA 2
EDA 3
⋮

⇓

Feature Engineering ⇒ Continuous process

⇓

Transformations

⇓

Algorithm Selection

⇓

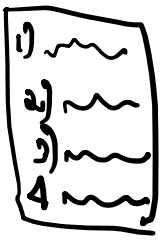
Hyperparameter Tuning

⇓

Model Evaluation

} Model

[Multiple versions of models / files / features / parameters]



Each step influences your final Model / your approach

You will try many different approaches / experiments

How do you keep a track ?

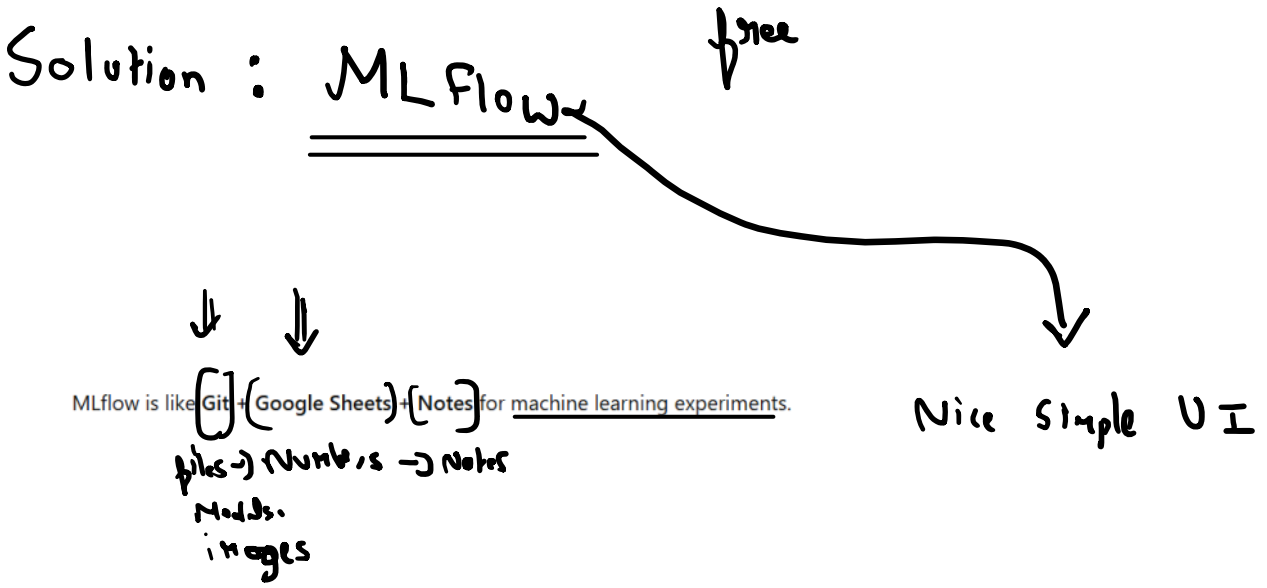
For example,

feature A, B, C with hyperparameter tuning of α, β, γ
gave accuracy of 92.1.

feature B, C with hyperparameter tuning of γ
gave accuracy of 83 %.

Modifying Model to Random Forest improved accuracy to 94 %.

It is difficult to keep track of these Manually



MLflow is like Git + Google Sheets + Notes for machine learning experiments.

files → Numbers → Notes
Models.
images

Typical situation where MLflow helps

You train:

- XGBoost with 10 parameter sets
- Random Forest with 5 variants
- Logistic Regression with feature tweaks

Without MLflow → confusion

With MLflow → clear winner, clean history, less stress

Model Governance

04 January 2026 18:00

ML system Design

AI Agent features Trigels

- 5$: 10%
- 2$: 90%

MC

Model governance in machine learning is the set of processes, controls, and accountability mechanisms that ensure ML models are **built, deployed, and used responsibly, reliably, and in line with business, legal, and ethical requirements.**

For Vishal what is the optimal price

Approaches ?

Prob of purchase
Clustering technique
optimal price

Vikram?
Teddy?

Evaluate

1. Model lifecycle control

Controls every stage:

- Problem definition
- Training & feature selection
- Validation
- Deployment
- Monitoring
- Retirement

Model
- TTV
- V
- Model evaluation
- Estimated profit

AB testing

=> 80%

2. Versioning & reproducibility

Ensures you can **recreate any model decision**.

Includes:

- Model versions
- Training data snapshot / hash
- Feature definitions
- Code + hyperparameters

Example

> "Why did the price prediction change on 12th Dec?"
> → Governance lets you trace it to **model v3.2 trained on data till Nov 30**

3. Validation & risk management

Formal checks before deployment:

- Performance (train vs validation vs OOT)
- Stability (PSI, CSI)
- Bias & fairness
- Stress testing

4. Explainability & transparency

Ability to explain:

- Global behavior (feature importance, PDPs)
- Local decisions (SHAP, LIME)

Governance requirement

> A model that can't be explained may be banned from high-risk use cases (credit, pricing, hiring).

5. Monitoring & drift detection

After deployment:

- Input drift
- Prediction drift
- Metric decay
- Data quality failures

Example

- Mileage distribution shifts → resale price model becomes unreliable
- Governance mandates retraining or rollback

Model Governance

Experimentation + Evaluation should be accounted for at all stages of your Model lifecycle

Auditing every version of your Model/approach/data.....

What does model governance involve in a machine learning project?
31 users have participated
A

Only deploying the best model for user access
8%
B

Ensuring that project code is well documented
6%
C

Tracking the experiment from the experimentation to deployment for auditing purposes
90%
D

Reducing the data required for training the model
0%

From <https://www.scaler.com/instructor/meetings/i/experiment-tracking-data-management-using-mlflow-53/>

Why is it important to keep track of data in machine learning projects?
32 users have participated
A

Data is the code that makes machine learning models work
6%
B

Data can be sourced from multiple storage units
6%
C

Data helps in analyzing features at different steps of the project
16%
D

All of the above
72%

From <https://www.scaler.com/instructor/meetings/i/experiment-tracking-data-management-using-mlflow-53/>

Our use case

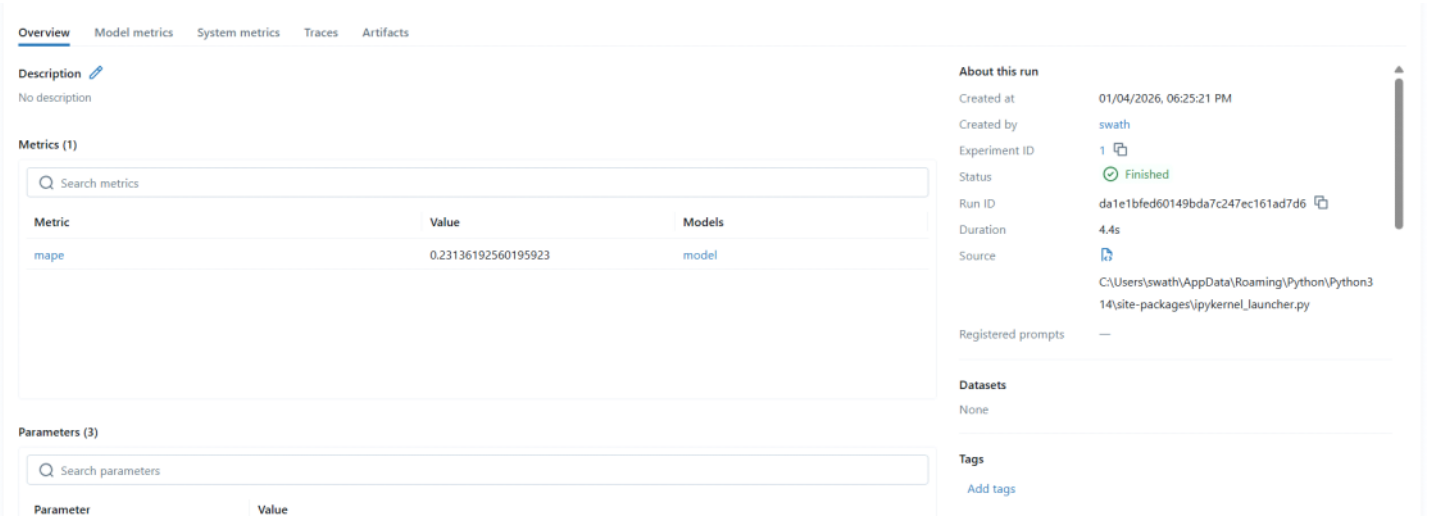
04 January 2026 12:53

Basic mlflow run

04 January 2026 12:53

Other important sections

04 January 2026 18:30



Traces

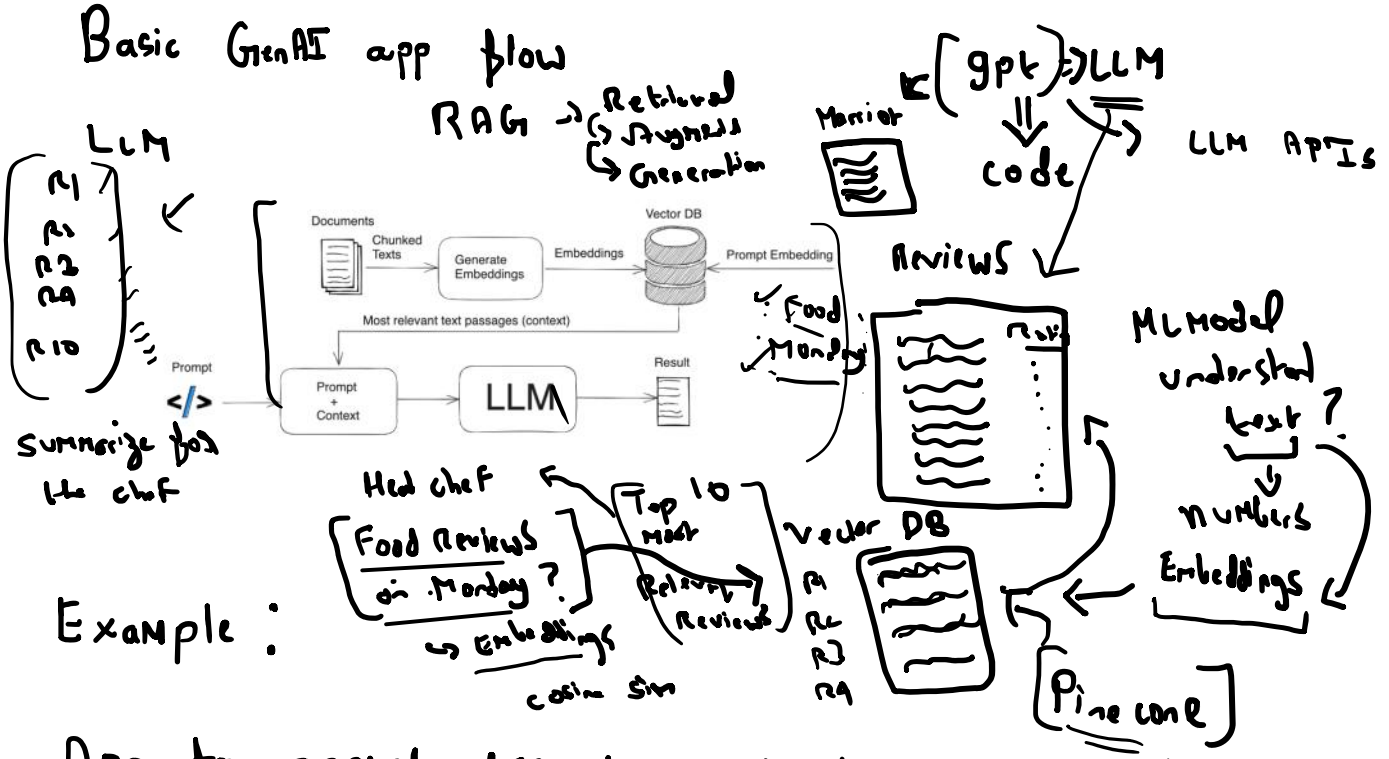
04 January 2026 18:34

One-line definition

Traces in MLflow record *step-by-step execution of an application*, not model training.

New feature introduced after emergence of GenAI (LLMOps)

Basic GenAI app flow



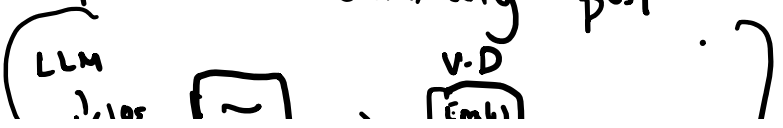
Example:

App to assist different departments in a popular hotel chain.

[Hundreds of customer reviews + ratings every month]

Based on reviews + ratings, give performance summary for:

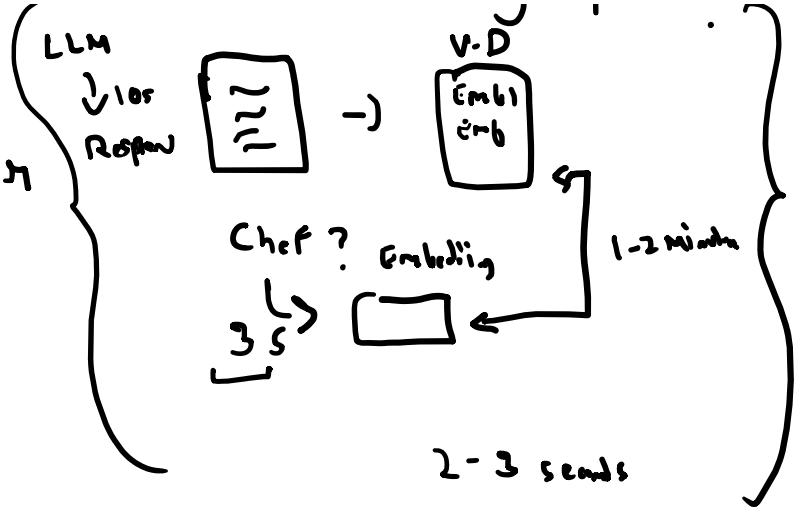
Food and Bev Manager



Food and Bev Manager

Housing and Accommodation Manager

Pool and Garden Manager



Overall summary for Hotel Manager

Role of traces :

Imagine an LLM app:

1. User query received
2. Vector search called
3. Prompt constructed
4. LLM called
5. Post-processing applied

A **trace** records:

- Each step
- Start / end time
- Inputs & outputs
- Errors (if any)

This is **debugging + observability**, not evaluation.

Why MLflow added traces

Classic ML:

- Train once
- Evaluate offline

LLM apps:

- Run continuously
- Many steps
- Hard to debug
- Latency matters

Why this is powerful (real problems it solves)

🔍 1. Debug bad answers

You can see:

- Did retrieval return junk?
- Was prompt malformed?

⏱️ 2. Latency optimization

You instantly know:

- LLM call = 1.8s bottleneck
- Retrieval = fine

Now you know **where to optimize**.